

Astrodynamics I: Final Exam

June 12, 2013 11:05AM-12:35PM (90 minutes)

Use the following constants and formulae without proof (if necessary).

- Radius of Earth $R_E = 6.378 \times 10^3 km = 2.09 \times 10^7 ft$
- Gravity Parameter of Earth $\mu_E = 3.986 \times 10^5 km^3 / sec^2 = 1.407 \times 10^{16} ft^3 / sec^2$
- $\vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right]$, $e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$

1. (15) Show that the specific energy is constant for a given orbit.
2. (20) A space object enters the atmosphere of the Earth at an altitude of 350,000ft with a velocity of 23,000ft/sec and a flight path angle of -30 degrees.
 - a. (6) What type of orbit does the space object follow?
 - b. (6) What were the velocity vector and flight path angle of the space object when it was at an altitude of 528,000ft?
 - c. (8) Ignoring the effect of the Earth's atmosphere on the motion of the object, determine if the space object impacts the surface of the Earth or not. If it does, what is its impact velocity vector? If not, what is its minimal miss altitude?
3. (15) Prove that the eccentricity vector always points to the periapsis.

To be continued on the next page

4. (15) Coordinate Transformations

- a. (3) Write down the rotation (coordinate transformation) matrix $R_x(\alpha)$, $R_y(\alpha)$ and $R_z(\alpha)$ about x, y and z-axis, respectively, for an arbitrary angle α .
- b. (12) Determine the following coordinate transformation matrices in terms of $R_x(\cdot)$, $R_y(\cdot)$ and $R_z(\cdot)$. It is NOT necessary to expand the products of matrices.
 - i. (4) From Geocentric Equatorial Coordinate System (IJK) to Topocentric Horizontal Coordinate System (SEZ)
 - ii. (4) From Perifocal Coordinate System (PQW) to Geocentric Equatorial Coordinate System (IJK)
 - iii. (4) From Geocentric Equatorial Coordinate System (IJK) to Satellite Coordinate System (NTW)

5. (20) Given the following set of orbital elements,

$$(a = 1.2175, e = 0.5195, i = 45^\circ, \Omega = 270^\circ, \omega = 104.82^\circ, \nu = 200.4418^\circ)$$

determine its position and velocity vector in the geocentric equatorial coordinate system. Use $\mu_E = 1DU^3 / TU^2$

6. (15) Use the flow chart discussed in class to briefly describe the coordinate transformation process between Celestial (=FK5=J2000) and Terrestrial (=ITRF) coordinates.